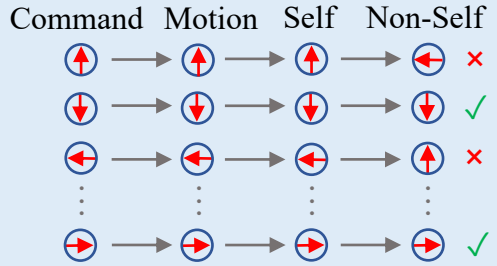
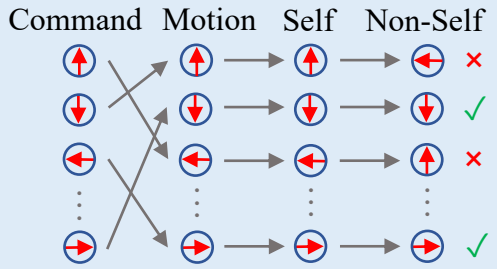
Scenario Design

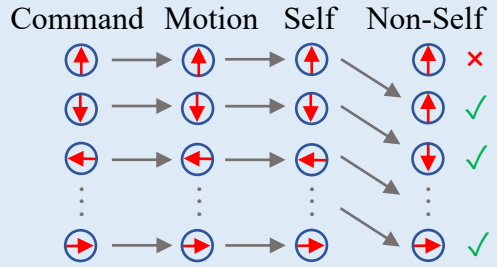
Scene 1: Same-step Causality
command-motion alignment

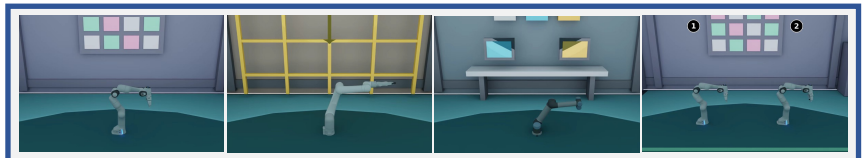


Scene 2: Mapping Stability
scrambled action space

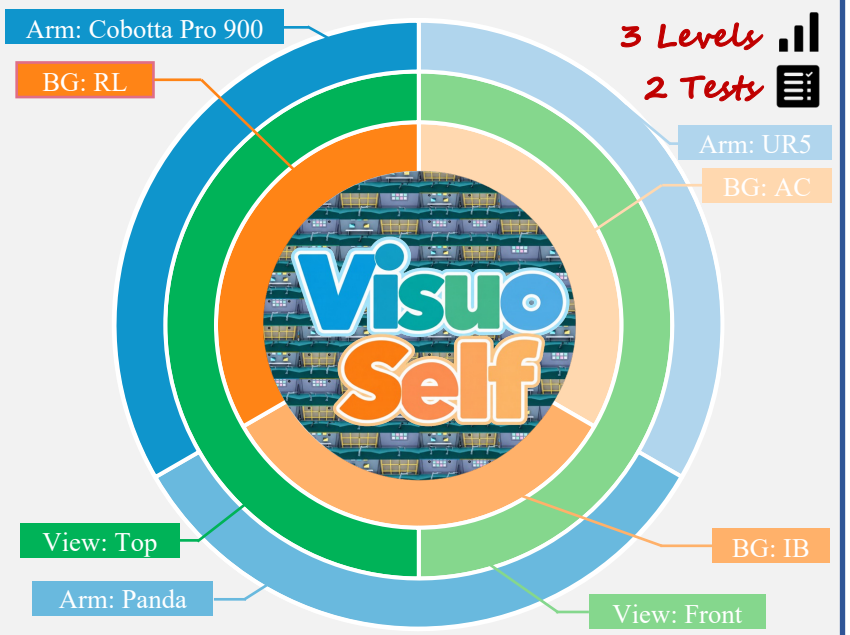


Scene 3: Delayed Causality
immediate vs. lagged response

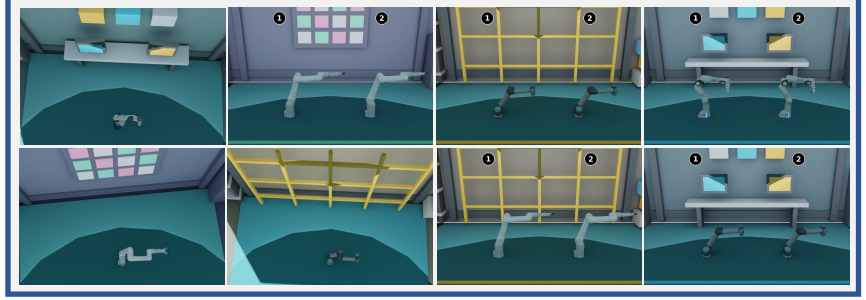




Dataset Contents

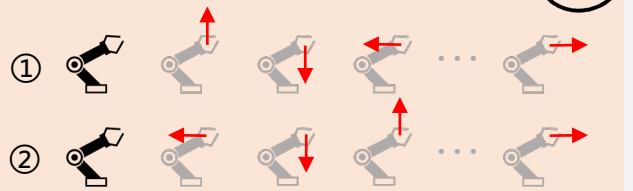


2500+ Episodes & 27000+ Steps for each MLLM

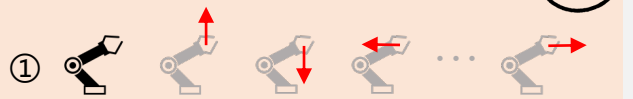


Evaluate Formats

Choice format
Choose from two candidate arms



Judgment format
Determine whether it is self



MLLMs can recognize the self when comparing candidates, yet perform poorly at identifying the self independently.

Logic Audit

Correct Answers | Wrong Answers

Structured Audit

- **Scene 1: step_match.** (match/mismatch)
- **Scene 2: cycle_stability** (same/different)
- **Scene 3: temporal_lag** (match/mismatch)

There is a significant gap between correctness at the option level and fine-grained behavioral verification.